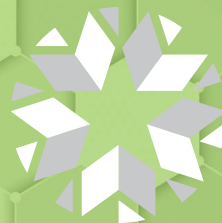


OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS
GRADE 5 SCIENCE



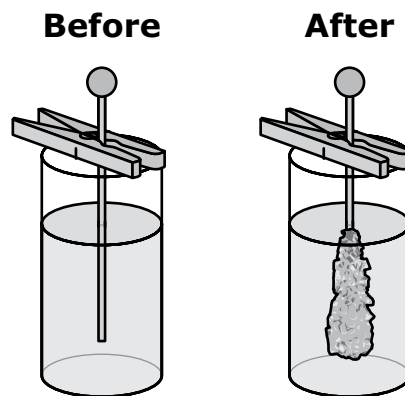
OKLAHOMA
Education

Sample Cluster for 5.PS1.1:

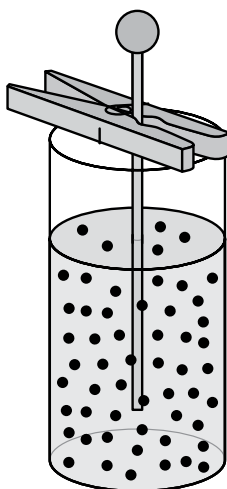
Use the information to answer the following questions.

A student and a teacher made homemade syrup from sugar and water. They noticed that when they added too much sugar to the water, the syrup formed crystals when it cooled. During an investigation of why this happened, the following occurred:

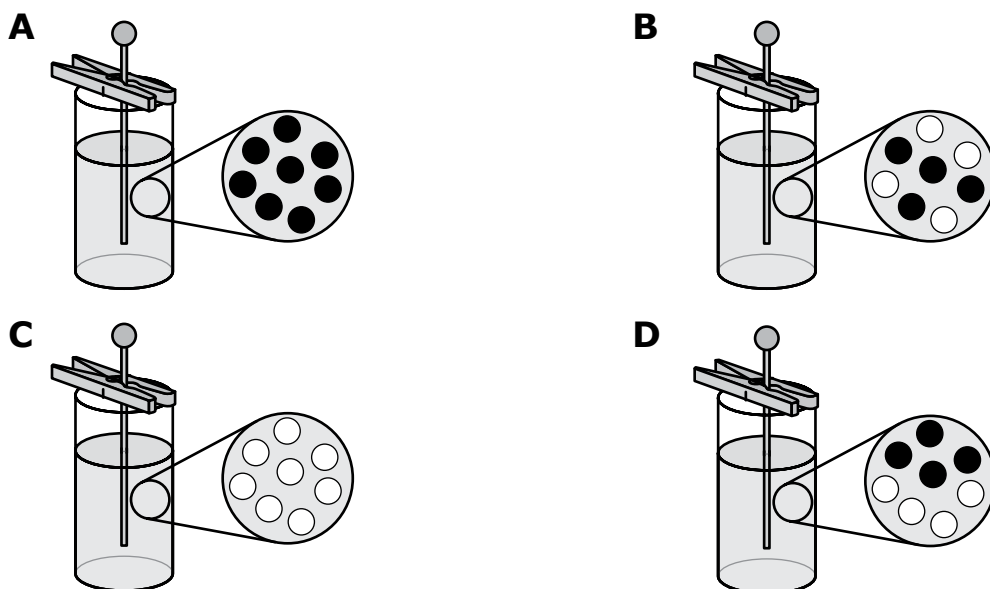
1. They added sugar to boiling water and stirred.
2. The sugar was no longer visible and the mixture was clear.
3. They poured the mixture into jars and placed a stick inside each jar.
4. After several days, crystals of sugar started forming around each stick.
5. The student drew these pictures of the jar before and after the crystals of sugar formed.



- 1** The teacher drew this model of the sugar and water particles after the sugar was added to the water. The teacher used black circles to represent sugar particles and white circles to represent water particles.



Which model shows the mixture in the jar when the sugar was no longer visible?



Standard: Develop a model to describe that matter is made of particles too small to be seen.

Cognitive Complexity:

SEP: Med

DCI: Med

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may not understand that sugar particles and water particles should be represented differently.
- B. KEY: There are different representations for sugar particles and water particles and the two types of particles are mixed together.**
- C. The student may not understand that sugar particles and water particles should be represented differently.
- D. The student may not understand that the sugar and water particles would not be divided into two separate groups; they should be mixed together.

- 2** Which statement **best** describes how sugar crystals formed during the investigation?
- A** It took a long time for sugar to form in water.
 - B** Sugar particles were destroyed when the water was stirred.
 - C** Sugar particles collected together until they were big enough to see.
 - D** The amount of sugar inside the jar increased when the stick was added.

Standard: Develop a model to describe that matter is made of particles too small to be seen.

Cognitive Complexity:

SEP: N/A

DCI: Low

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may not understand that the amount of time does not describe how the sugar crystals formed.
- B. The student may think that dissolving destroys matter.
- C. **KEY: While dissolved in water, sugar particles are too small to be seen; as sugar particles come out of solution, the particles grow in size until they are visible.**
- D. The student may think that so much sugar was added that sugar particles were visible.

- 3** What should a model show to **best** explain how the sugar became no longer visible?
- A** sugar collecting
 - B** sugar evaporating
 - C** sugar changing into water
 - D** sugar breaking into pieces

Standard: Develop a model to describe that matter is made of particles too small to be seen.

Cognitive Complexity:

SEP: High

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Low Support

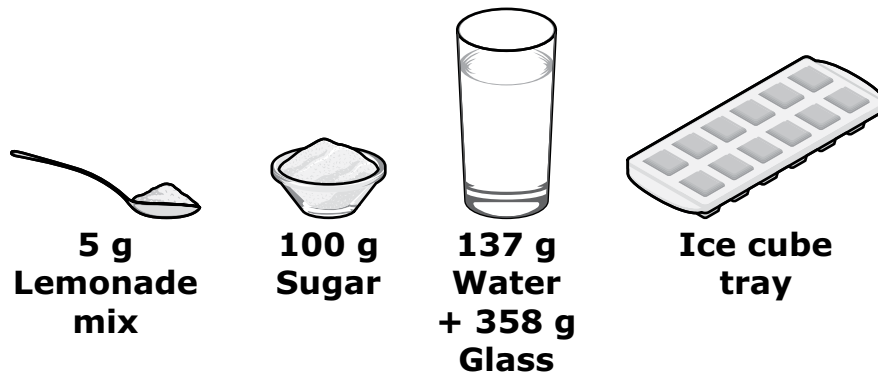
Rationale:

- A. The sugar collecting together would be larger and easier to see.
- B. The sugar doesn't evaporate.
- C. The sugar doesn't become water.
- D. **KEY: The sugar breaks into particles too small to be seen.**

Sample Cluster for 5.PS1.2:

Use the information to answer the following questions.

Two students investigated what happens when matter changes form. The materials the students used are shown in the pictures. The students used the amounts of lemonade mix, sugar, and water shown.



Then the students followed this procedure.

1. Make lemonade from the lemonade mix, sugar, and water.
2. Pour all of the lemonade into the ice cube tray. Put the same amount of lemonade into each spot in the tray. Leave no lemonade left over. Cover the tray and place it in the freezer overnight.
3. Remove the ice cube tray from the freezer the next day. See that the liquid lemonade has frozen into lemon ice. See that the cubes of lemon ice are taller than the sides of the tray.

- 4** The students measured each material before stirring them together. The students slowly stirred the lemonade mix and sugar into the glass of water. The lemonade mix and sugar both seemed to disappear. Several students thought the amount of matter changed when the lemonade mix and sugar disappeared. The students decide to measure the mixture to find out if the amount of matter changed.

Which statement best explains which tool should be used to show what happens to the amount of matter after the mixture is stirred?

- A** The students should use a scale which will show that the total weight stayed the same.
- B** The students should use a measuring cup which will show that the total volume stayed the same.
- C** The students should use a scale which will show that the total weight decreased.
- D** The students should use a measuring cup which will show that the total volume decreased.

Standard: Measure and graph quantities to provide evidence that regardless of the type of change that occurs when heating, cooling, or mixing substances, the total weight of matter is conserved.

Cognitive Complexity:

SEP: N/A

DCI: Med

CCC: High

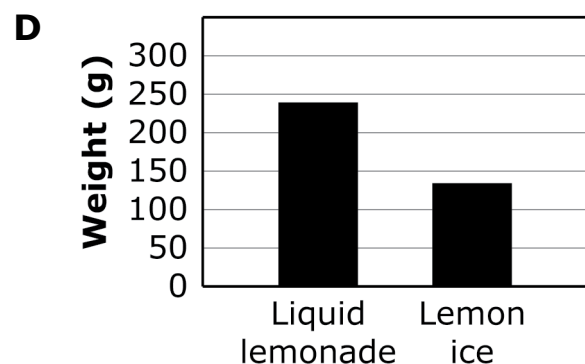
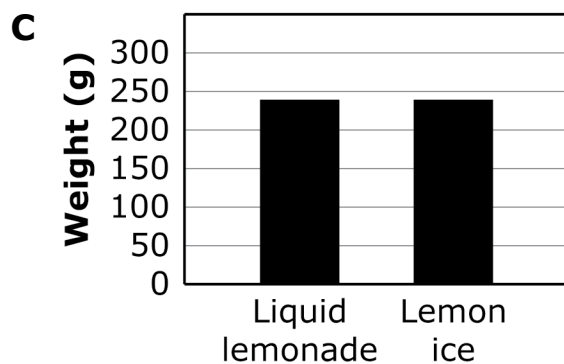
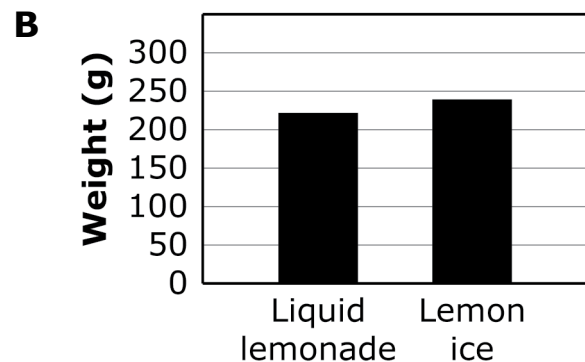
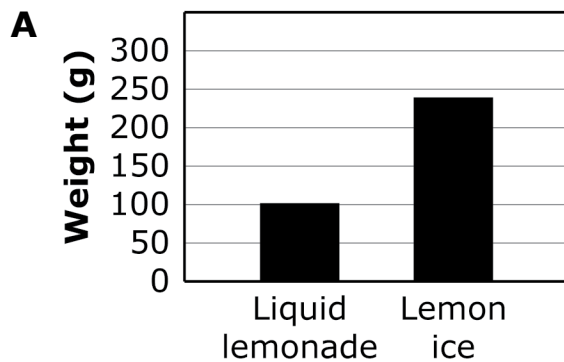
Overall Cognitive Complexity: Moderate Support

Rationale:

- A. KEY:** The amount of matter is measured using mass/weight. Matter is conserved so the weight will remain the same.
- B.** The student may have thought that matter was measured using volume.
- C.** The student may have thought that the amount of matter decreased when the substances dissolved.
- D.** The student may have thought that matter was measured using volume.

- 5** The students removed the lemon ice from the ice cube tray at the end of the investigation. Then they measured the total weight of all the lemon ice cubes.

Which graph shows the total weight of the liquid lemonade before it was poured into the tray and the total weight of the lemon ice removed from the tray?



Standard: Measure and graph quantities to provide evidence that regardless of the type of change that occurs when heating, cooling, or mixing substances, the total weight of matter is conserved.

Cognitive Complexity:

SEP: Med

DCI: Med

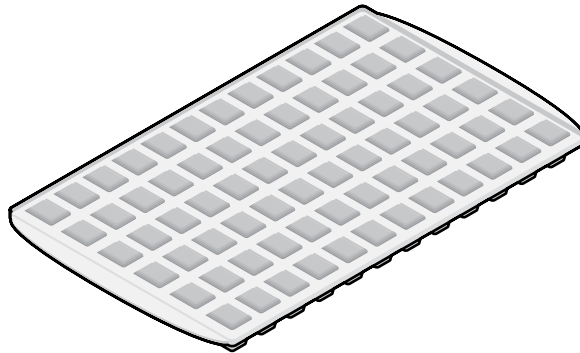
CCC: Med

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may think that the mass of the water and the lemonade mix is not included in the liquid lemonade or that the mass of lemon ice is greater because its size is greater.
- B. The student may think that the mass of the lemonade mix is not included in the liquid lemonade or that the mass of lemon ice is greater because its size is greater.
- C. **KEY: Freezing lemonade does not change the mass.**
- D. The student may think that the mass of the lemonade mix and the sugar is not included in the lemon ice.

- 6** The students decided to let the lemon ice melt after the investigation. Once the lemon ice melted, the students poured all the liquid into a different ice cube tray. The drawing below shows this new tray.



The students poured the same amount of lemonade into each spot in the tray. There was no lemonade left over. The students covered the tray and placed it in the freezer overnight. The students removed the tray from the freezer the next day.

Which statement is correct about the new lemon ice cubes?

- A** The new lemon ice weighed less than the lemon ice made the first time.
- B** Each new lemon ice cube had the same weight as each lemon ice cube made the first time.
- C** Each new lemon ice cube contained more matter than each lemon ice cube made the first time.
- D** The new lemon ice contained the same amount of matter as the lemon ice made the first time.

Standard: Measure and graph quantities to provide evidence that regardless of the type of change that occurs when heating, cooling, or mixing substances, the total weight of matter is conserved.

Cognitive Complexity:

SEP: N/A

DCI: High

CCC: High

Overall Cognitive Complexity: Low Support

Rationale:

- A. The student may think that smaller ice cubes result in less total mass.
- B. The student may confuse the conservation of mass with the mass of the different sized ice cubes.
- C. The student may think that more ice cubes results in each having more mass.
- D. KEY: The total amount of matter stays the same throughout these steps.**

Use the information to answer the following questions.

A student wanted to make cookies. They poured flour, sugar, powdered sugar, baking soda, and salt into five separate bowls. When it was time to measure the substances, the student forgot which one was in each bowl. To decide which substance was in each bowl, the student researched the substances’ properties. They made a table based on their research.

Properties of White Baking Substances

	Baking Soda	Flour	Sugar	Powdered Sugar	Salt
Appearance	Powder	Powder	Crystals	Powder	Crystals
Change when Heated	None	Yellow	Brown	Brown	None
Bubbles in Vinegar	Yes	No	No	No	No
Dissolves in Water	Yes	No	Yes	Yes	Yes

(Items on the following pages)

7 The student carried out three tests with a white baking substance. They mixed the substance with water. Then they mixed the substance with vinegar. Finally, they heated the substance. The student recorded the observations shown.

- Small crystals
- Dissolved in water
- No bubbles in vinegar
- No change when heated

Which substance did the student have?

- A** Baking soda
- B** Flour
- C** Sugar
- D** Salt

Standard: Make observations and measurements to identify materials based on their properties.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. Baking soda does not change when heated and it does bubble in vinegar.
- B. Flour does not dissolve in water and does not look like small crystals.
- C. Granulated sugar does melt.
- D. **KEY:** Salt is the only substance that exhibits all four properties the student observed for the white powder.

8 The student compared the two sugars.

Which statement explains how the student could tell the difference between the two types of sugars?

- A** by stirring both sugars into a cup of water
- B** by heating both sugars
- C** by viewing both sugars under a hand lens
- D** by adding both sugars to vinegar

Standard: Make observations and measurements to identify materials based on their properties.

Cognitive Complexity:

SEP: Low

DCI: Low

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. This will not allow the student to differentiate between the two sugars because both sugars dissolve in water.
- B. The properties the student will examine will differentiate the sugars from the other powders but not from each other as they both turn brown when heated.
- C. KEY: The hand lens will allow the student to see that one sugar is made up of crystals and the other of powder.**
- D. Neither sugar bubbles in vinegar and therefore the student will not be able to differentiate between the two types of sugar.

- 9** The student had a substance that did not bubble in vinegar and wanted to see if they could figure out what the substance was by only using heat. The student noticed that some of the substances have multiple reactions when heated.

Which statement explains if the student was able to tell what substance they had from using only the two properties?

- A** The student will be able to tell the difference because powder sugar turns brown when heated.
- B** The student will be able to tell the difference because salt and sugar are both crystal in appearance.
- C** The student will not be able to tell the difference because flour is powder in appearance and sugar is crystal in appearance.
- D** The student will not be able to tell the difference because both sugars turn brown when heated.

Standard: Make observations and measurements to identify materials based on their properties.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

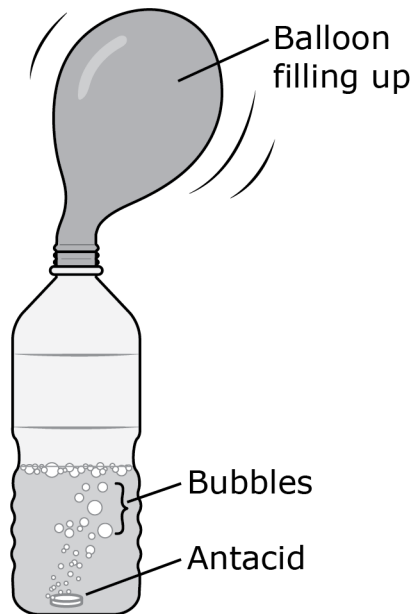
Overall Cognitive Complexity: Moderate Support

Rationale:

- A. Both sugars turn brown when heated, therefore if the substance the student is testing is sugar, they will not be able to tell the difference.
- B. The student is using the wrong property to determine what the substance is.
- C. The student is using the wrong property to determine what the substance is.
- D. **KEY:** The student will not be able to tell what the substance is by only using the two properties because both sugars turn brown when heated and do not bubble in vinegar.

Use the information to answer the following questions.

Students watched their teacher drop an antacid into a bottle of water and mix the substances. A balloon was placed onto the opening of the bottle. The students noticed that after the substances were mixed, bubbles were present in the water and the balloon filled up so that it became bigger. The diagram shows what the students observed.



The students wonder why this happened and decide to mix different substances to investigate. The teacher gives the students several substances for their investigation:

- An antacid (a white solid)
- Water (a clear liquid)
- Lemon juice (a light yellow liquid)
- Salt (a white solid)

(continued on next page)

The students make three different mixtures using these substances and record their observations in a table.

Investigation Observations After Mixing

Observations After Mixing	Mixture 1 (Antacid + Water)	Mixture 2 (Antacid + Lemon Juice)	Mixture 3 (Lemon Juice + Salt)
Presence of bubbles	Yes	Yes	No
Temperature change	Yes	Yes	No
Solid is left over	No	No	Yes

10 The students claim that a new substance formed in Mixture 2.

Which observation from the students' investigation supports this claim?

- A** The mixture is a liquid.
- B** The mixture has lemon juice.
- C** Bubbles are present in the mixture.
- D** Solids are not present in the mixture.

Standard: Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

Cognitive Complexity:

SEP: Low

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Low Support

Rationale:

- A. The state of the mixture is irrelevant to whether a new substance formed.
- B. This information about what substances were combined to form the mixture does not indicate whether a new substance formed.
- C. KEY: Bubbles indicate that gas formed, which would support the students' claim that a new substance formed.**
- D. This observation would not support the students' claim because it would indicate that a new solid substance did not form.

11 The students make a fourth mixture by mixing water and an unknown solid in a bottle and placing a balloon on the opening of the bottle.

Which observation would **best** help the students determine that a new substance may have been formed in Mixture 4?

- A** After mixing, the solid dissolved in water.
- B** After mixing, the balloon remained unfilled.
- C** After mixing, the resulting liquid looked clear.
- D** After mixing, there was a temperature change.

Standard: Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Low Support

Rationale:

- A. A solid can dissolve in a liquid without forming a new substance.
- B. This evidence would suggest that a gas did not form to fill the balloon, which would indicate that a new substance did not form.
- C. Substances can change appearance in mixtures without a new substance forming.
- D. KEY: A change in temperature can provide evidence that a chemical reaction occurred and therefore a new substance formed.**

- 12** The students want to change their investigation to gather evidence that a new substance formed.

Which change to their investigation procedure would **best** help the students gather evidence of a new substance forming?

- A** Mix different amounts of each substance in each mixture.
- B** Increase the amount of each substance in each mixture.
- C** Determine whether some of the substances remain solid after becoming part of each mixture.
- D** Record whether there is a difference between the properties of each substance and the mixture.

Standard: Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

Cognitive Complexity:

SEP: High

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Low Support

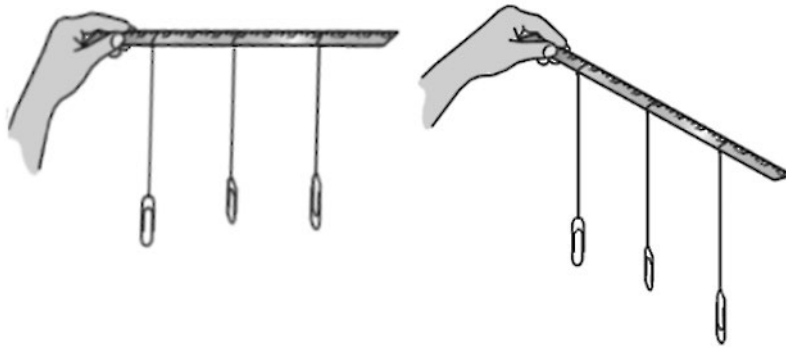
Rationale:

- A. The amount of each substance mixed is irrelevant to whether new substances form afterward and the variable of how much each substance is mixed would not be controlled to gather reliable evidence.
- B. Changing the amount of substance mixed would not affect the outcome of the same substances being combined together and may not result in a new substance forming.
- C. Students may have the misconception that after a chemical reaction, the product is a mixture in which some of the substances remain unchanged.
- D. KEY: Measuring whether there was a change in properties could indicate whether a new substance formed as a result of substances being mixed.**

Use the information to answer the following questions.

A student was outside during recess and noticed leaves falling from a tree. The student wondered why the leaves fell down. Their teacher decided to have the students conduct their own experiments to observe for themselves how gravity affects objects. A group of students did an experiment using paperclips and a ruler. First, the students tied metal paper clips to a metal ruler using a string. The drawing shows what happened when the students tilted the ruler.

Experiment



13 Which observation during the experiment **best** supports the claim that gravity pulls objects toward the center of Earth?

- A** The paper clips swing freely when the ruler is tilted.
- B** The paper clips are all the same distance from the tilted ruler.
- C** The paper clips hang straight down no matter how the ruler is tilted.
- D** The paper clips remain attached to the ruler no matter how the ruler is tilted.

Standard: Support an argument, with evidence, that Earth's gravitational force pulls objects downward toward the center of the Earth.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The swinging doesn't indicate that the gravitational force pulls objects toward the center of Earth.
- B. The distance, which is directed by the string, doesn't indicate that the gravitational force pulls objects toward the center of Earth.
- C. KEY: This shows that gravity pulls the object straight down toward Earth's center.**
- D. The attachment doesn't indicate that the gravitational force pulls objects toward the center of Earth.

- 14** How could the experiment be revised to best provide additional evidence to support the claim that Earth's gravity pulls objects toward the center of Earth?
- A** Repeat the experiment with a wooden ruler.
 - B** Repeat the experiment outside on a windy day.
 - C** Repeat the experiment in different places on Earth.
 - D** Repeat the experiment on different planets in our solar system.

Standard: Support an argument, with evidence, that Earth's gravitational force pulls objects downward toward the center of the Earth.

Cognitive Complexity:

SEP: Medium

DCI: Low

CCC: Medium

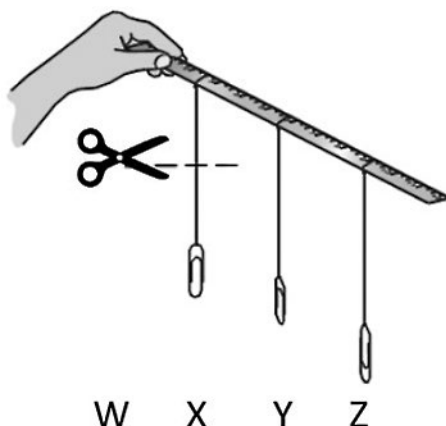
Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The material the ruler is made out of doesn't affect gravity.
- B. This could affect the strings hanging straight down if the wind blows them.
- C. KEY:** If the experiment shows that everywhere on Earth objects fall toward the ground, then that lends support to the claim that gravity pulls objects toward the center of the Earth.
- D. Repeating the experiment on other planets will show the effect of those planets' gravity on the paper clips.

15 The students cut the string on the first paperclip from the tilted ruler.

Select the location where the paper clip will fall when the string is cut and provide the **best** explanation for your selection.



- A** Position Z because the ruler is tilted
- B** Position X because the Earth's gravity pulls objects downward towards the center of the Earth
- C** Position Y because the ruler is tilted
- D** Position W because the Earth's gravity pulls objects downward towards the center of the Earth

Standard: Support an argument, with evidence, that Earth's gravitational force pulls objects downward toward the center of the Earth.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

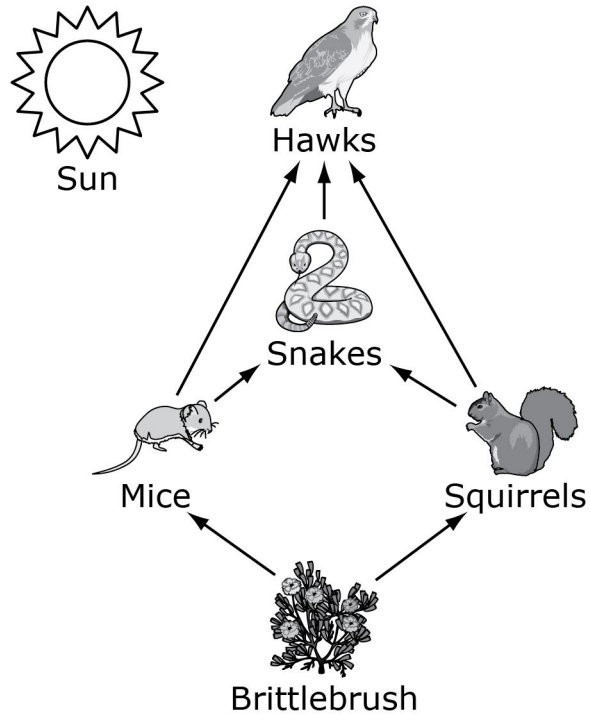
- A. The ruler being tilted will not affect where the paper clip falls on the ground.
- B. KEY: The paper clip will fall straight down to Position X because Earth's gravity pulls objects downward to the center of the Earth.**
- C. The ruler being tilted will not affect where the paper clip falls on the ground.
- D. The paper clips will not fall to Position W because it is not straight below the paper clip that has the string being cut.

Sample Cluster for 5.PS3.1:

Use the information to answer the following questions.

A student is learning about the desert ecosystem in class. They learn that even though the Sun makes the desert hot, many different organisms can survive there. The student is given a food web of a desert ecosystem. The food web includes brittlebrush, which is a type of plant.

Desert Ecosystem Food Web



(Items on the following pages)

- 16** Based on the Desert Ecosystem Food Web, which statement describes the movement of energy in this ecosystem?
- A** Snakes get energy from squirrels and provide it to hawks.
 - B** Hawks get energy from snakes and provide it to squirrels.
 - C** Brittlebrush gets energy from mice and squirrels and provides it to hawks.
 - D** Squirrels get energy from snakes and hawks and provide it to brittlebrush.

Standard: Use models to describe that energy in animals' food (used for body repair, growth, motion, and to maintain body warmth) was once energy from the sun.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. KEY:** The food web indicates that snakes eat squirrels and hawks eat snakes, so energy is transferred from squirrels to snakes to hawks.
- B.** The food web indicates that hawks eat squirrels, so energy is transferred from squirrels to hawks.
- C.** The food web indicates that mice and squirrels eat brittlebrush, and hawks eat mice and squirrels, so energy is transferred from brittlebrush to mice and squirrels to hawks.
- D.** The food web indicates that snakes eat mice and mice eat brittlebrush, so energy is transferred from brittlebrush to mice to snakes to hawks.

- 17** Which statement about the Desert Ecosystem Food Web **best** describes the original source of energy for hawks in this ecosystem?
- A** brittlebrush, because hawks get energy directly from brittlebrush
 - B** mice, because hawks eat mice and this food helps hawks grow
 - C** the soil, because brittlebrush gets energy from the soil and hawks gain that energy when they eat brittlebrush
 - D** the Sun, because brittlebrush uses energy from the Sun to grow and hawks gain that energy when they eat mice

Standard: Use models to describe that energy in animals' food (used for body repair, growth, motion, and to maintain body warmth) was once energy from the sun.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. Plants like brittlebrush derive their energy from the Sun and hawks do not eat brittlebrush, but instead eat the mice and squirrels that eat brittlebrush.
- B. Mice are not the original source of energy in this ecosystem since energy is transferred from the Sun to brittlebrush plants and then to mice.
- C. Students may have the misconception that brittlebrush plants get their energy from the soil rather than the Sun and that hawks directly eat brittlebrush instead of eating the mice and squirrels that eat brittlebrush plants.
- D. **KEY: All organisms in the food web either directly or indirectly eat plants, like brittlebrush, and plants obtain their energy from the Sun.**

18 The student learns that while deserts can be very hot during the day, they can be very cold at night. While thinking about this information, the student wonders how squirrels would be affected at night if there were no brittlebrush plants in the desert.

Based on the Desert Ecosystem Food Web, which statement **best** describes how squirrels would be affected in a desert ecosystem without brittlebrush plants?

- A** Squirrels would get energy they need from the air to stay warm on cold nights.
- B** Squirrels would not need the energy they get from hawks to stay warm on cold nights.
- C** Squirrels would not get the energy from food that is needed to stay warm on cold nights.
- D** Squirrels would need to make their own energy from sunlight to stay warm on cold nights.

Standard: Use models to describe that energy in animals' food (used for body repair, growth, motion, and to maintain body warmth) was once energy from the sun.

Cognitive Complexity:

SEP: High

DCI: High

CCC: High

Overall Cognitive Complexity: Low Support

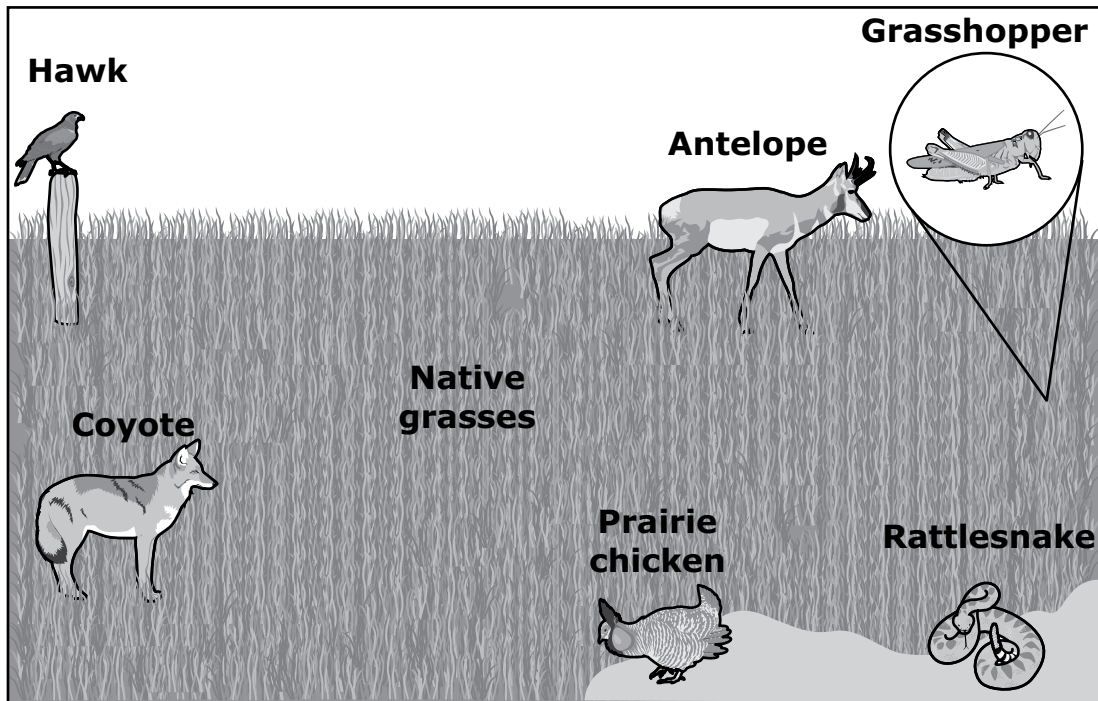
Rationale:

- A. Squirrels do not obtain energy from the air but instead from eating brittlebrush plants, and that food provides squirrels with the energy they need to maintain body warmth.
- B. Squirrels eat brittlebrush plants, not hawks.
- C. KEY: The food web indicates that energy is transferred from brittlebrush plants to squirrels and the food that squirrels eat provides them with the energy needed to maintain body warmth.**
- D. Students may have the misconception that squirrels can make their own food like plants.

Sample Cluster for 5.LS2.1:

Use the information to answer the following questions.

The drawing shows some plants and animals that live in the Black Kettle National Grassland in southwestern Oklahoma.



Some students wanted to make a model to show how matter moves through this grassland. The students had learned that the movement of matter allows plants and animals in the grassland to get nutrients or food. If the plants and animals do not get the nutrients or food they need, they cannot survive.

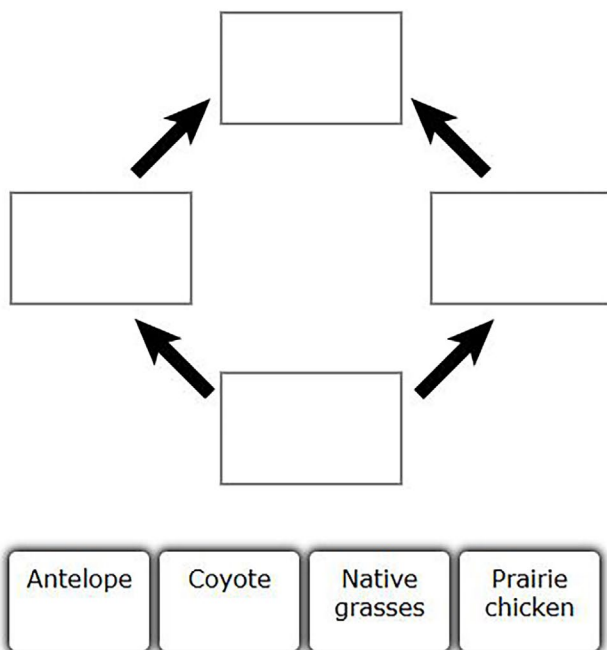
By making the model, the students could predict how well plants and animals would survive if events such as fire or drought happened in the ecosystem.

(Items on the following pages)

The students also learned what some of the organisms eat.

- Prairie chickens eat native grasses and coyotes eat prairie chickens.
- Native grasses are eaten by antelope and antelope are eaten by coyotes.

Drag the organisms into the food web to show how matter moves among the organisms. To drag an organism, click and hold the organism, and then drag it to the desired space. To change an organism, click and hold it, and then drag it back to the original location.



Standard: Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Cognitive Complexity:

SEP: Low

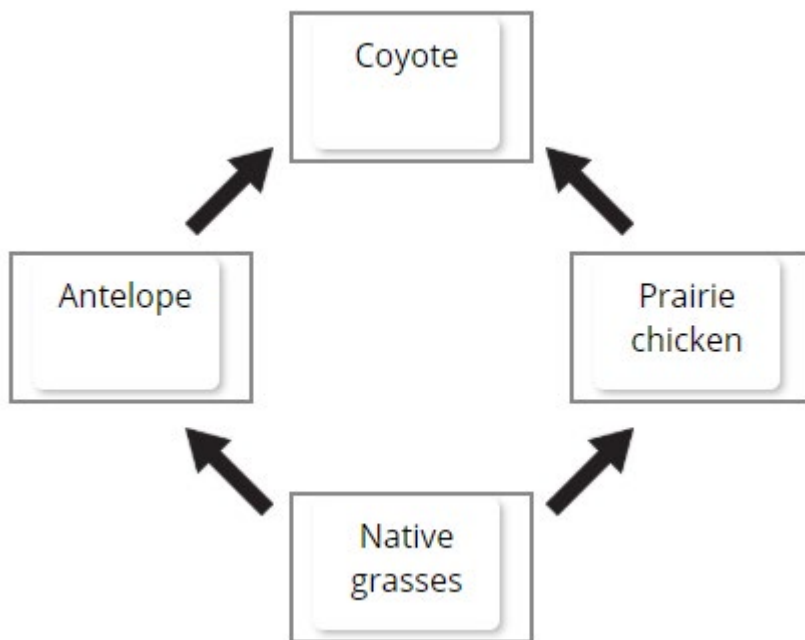
DCI: Low

CCC: Low

Overall Cognitive Complexity: Scripted

Scoring:**Rubric**

Score	Description
2	Students place all four organisms in the correct place.
1	Students place two or three organisms in the correct place.
0	Students place one or no organisms in the correct place.
Blank	No response

KEY:**Rationale:**

The arrows indicate that matter moves from native grasses to antelope and prairie chickens and then moves from antelopes and prairie chickens to coyotes.

20 What can the students add to their model to show that matter also moves between organisms and the environment?

- A** rocks, because they are a common part of soil
- B** Sun, because it allows plants to make their own food
- C** wind, because it moves air and dust around the grassland
- D** decomposers, because they break down dead plants and animals

Standard: Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Cognitive Complexity:

SEP: Low

DCI: Low

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may think that rocks in the soil will show movement of matter.
- B. The student may think that the Sun provides matter for organisms.
- C. The student may think that wind will show movement of matter into organisms.
- D. KEY: Decomposers (an organism) move matter from organisms into the environment.**

- 21** Which set of events should the students' model also include to show how matter moves in the ecosystem?
- A** Plants take up air and water to make food. → Animals eat plants. → Animals breathe out air.
 - B** Plants release food as waste. → Animals break down waste from plants. → Animals breathe out air.
 - C** Animals take in air and water to make food. → Other animals eat these animals. → Animals release waste.
 - D** Animals release waste into air. → Water is taken up by animals to make food. → Other animals eat these animals.

Standard: Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Cognitive Complexity:

SEP: Med

DCI: Med

CCC: Med

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. KEY: This traces part of the path carbon takes in an ecosystem.**
- B. The student may think that plants make food as waste and that animals break down waste from plants.
- C. The student may think that animals make food.
- D. The student may think that animals make food and that animals exhale water.

Use the information to answer the following questions.

constellation—a group of stars

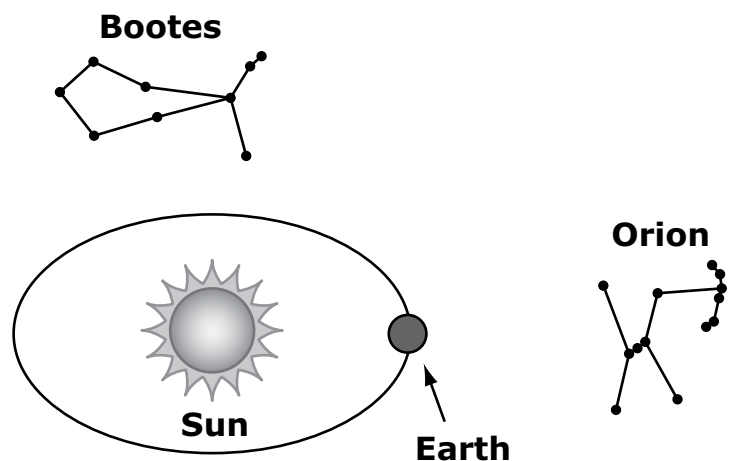
A student in Oklahoma studying the night sky wondered why different stars are seen at different times of the year. The student decided to study two constellations. One of the constellations was Orion, and the other constellation was Bootes.

The student found two pieces of information about the constellations. First, the student found the number of hours Orion is visible each night. The student recorded the data for different months in a table. Next, the student found a picture to show where Earth is in its path around the Sun in December. The student copied the picture and also marked where the constellations, Orion and Bootes, are in December.

The student's table and picture are shown.

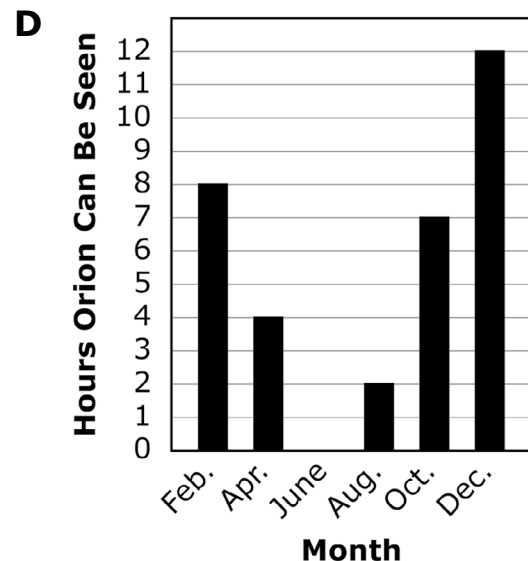
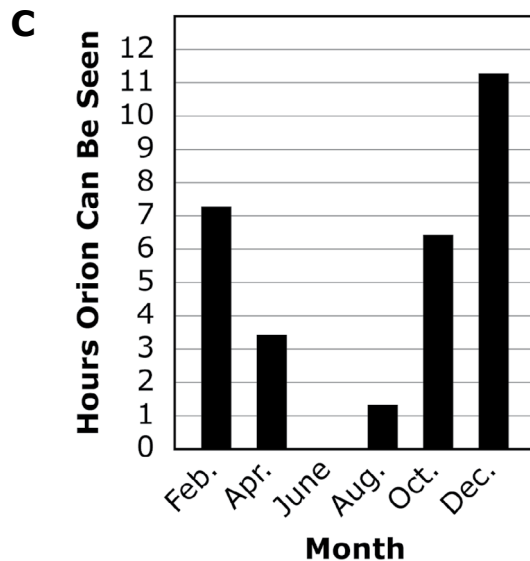
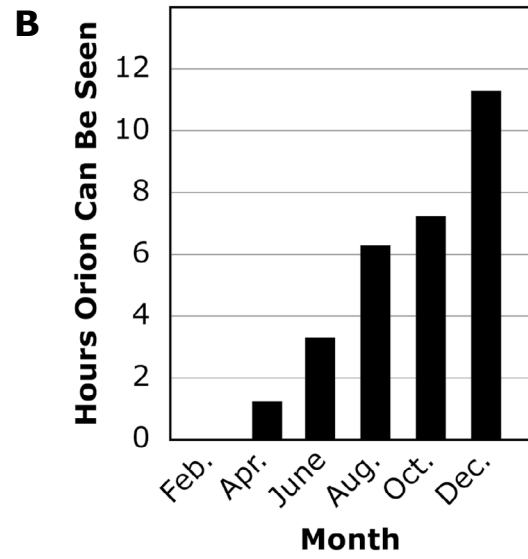
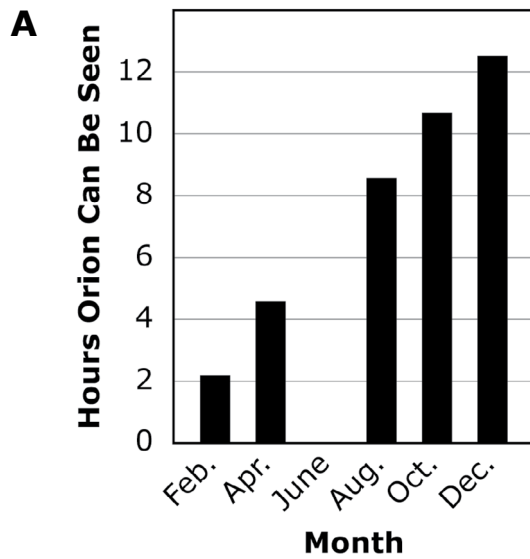
When Can Orion Be Seen?

Month	Hours Orion Can Be Seen at Night in Oklahoma
February	7.2
April	3.4
June	0.0
August	1.3
October	6.4
December	11.2



(Items on the following pages)

- 22** During the year, the number of hours Orion can be seen in Oklahoma changes. Which graph correctly shows the changes?



Standard: Represent data in graphical displays to reveal patterns of daily changes in length and direction of shadows, in addition to different positions of the sun, moon, and stars at different times of the day, month, and year.

Cognitive Complexity:

SEP: Low

DCI: N/A

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may think the values should increase from left to right and incorrectly interprets some of the numbers in the data table.
- B. The student may think the values should increase from left to right and that the values in the graph should not match what is in the data table.
- C. **KEY:** This graph has a proper y-axis and correctly matches the data from the table to the data shown in the graph.
- D. The student may think that they should round the values in the graph to the nearest integer.

23 Some months are missing from the student's data table.

Which table shows the number of hours Orion will likely be seen in September and November in Oklahoma?

A

Month	Hours Orion Can Be Seen at Night
September	1.25
November	6.25

B

Month	Hours Orion Can Be Seen at Night
September	3.8
November	9.8

C

Month	Hours Orion Can Be Seen at Night
September	0.8
November	5.3

D

Month	Hours Orion Can Be Seen at Night
September	7.25
November	3.25

Standard: Represent data in graphical displays to reveal patterns of daily changes in length and direction of shadows, in addition to different positions of the sun, moon, and stars at different times of the day, month, and year.

Cognitive Complexity:

SEP: High

DCI: Low

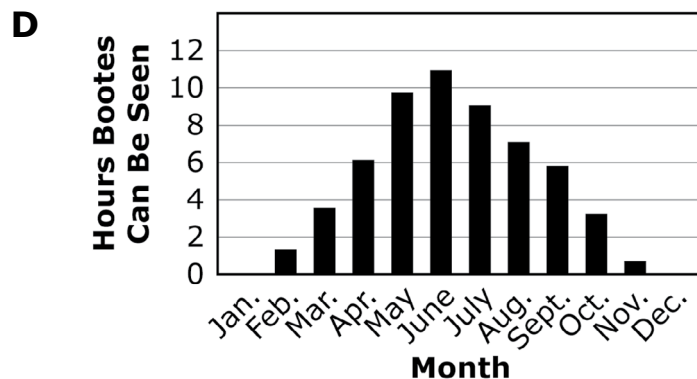
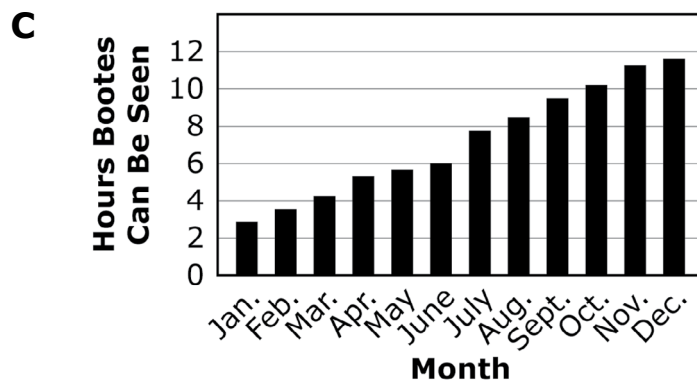
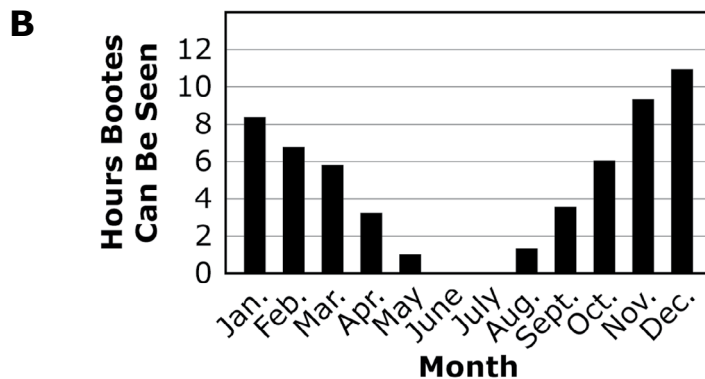
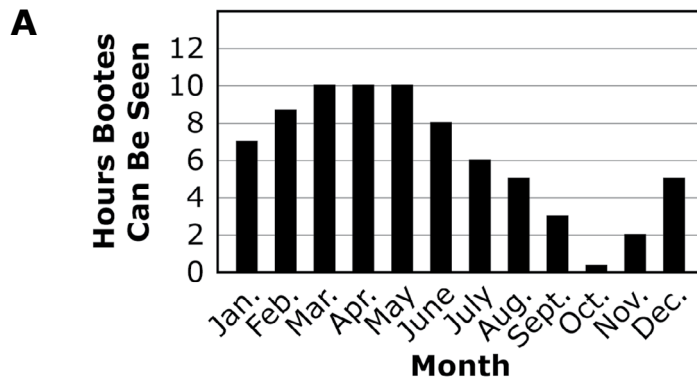
CCC: High

Overall Cognitive Complexity: Low Support

Rationale:

- A. The student may think that the time Orion is visible each month after the months shown in the table should be less than the previous month.
- B. **KEY:** The September value is between the August and October values while the November value is between the October and December values.
- C. The student may think that the time Orion is visible each month after the months shown in the table should be significantly less than the previous month.
- D. The student may think that September is a month that Orion is visible for a peak amount of time before decreasing into October and November.

- 24 Which graph shows the number of hours Bootes will **most likely** be seen in the Oklahoma night sky during the year?



Standard: Represent data in graphical displays to reveal patterns of daily changes in length and direction of shadows, in addition to different positions of the sun, moon, and stars at different times of the day, month, and year.

Cognitive Complexity:

SEP: High

DCI: Low

CCC: High

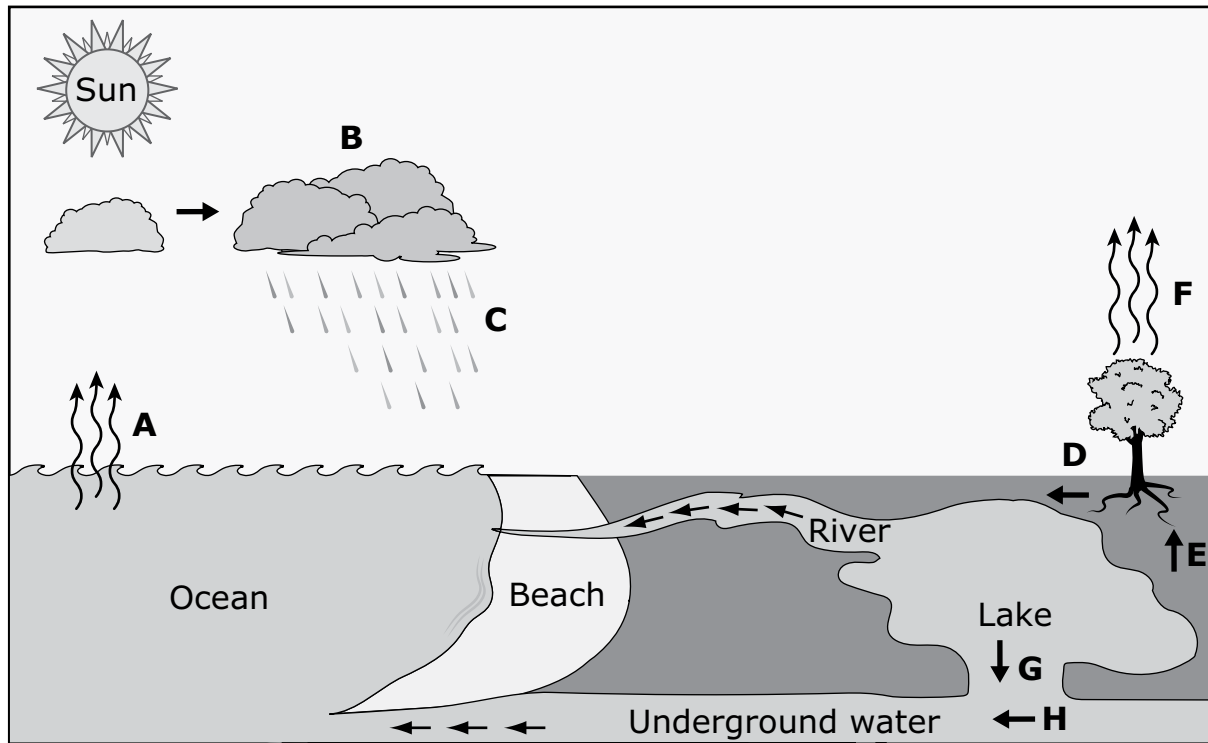
Overall Cognitive Complexity: Low Support

Rationale:

- A. **KEY:** Given the location of Bootes relative to Orion and the Orion data, this graph shows the amount of time Bootes will be visible each month.
- B. The student may think that Bootes will have the same amount of visible time as Orion.
- C. The student may think that Bootes has an increasing visibility time throughout a calendar year.
- D. The student may think that Bootes will be offset 6 months relative to Orion, rather than 3 months.

Use the information to answer the following questions.

A student watched a news show about a large amount of rainfall one summer on a tropical island. To better understand where all the water in the air came from, he looked at this water cycle model.



Key

- A. ?
- B. Water in the air changes state to form clouds.
- C. Water falls back to Earth's surface as rain, snow, and other forms.
- D. Water moves across Earth's surface in streams and rivers that flow into lakes and oceans.
- E. Plant roots absorb water from the ground.
- F. ?
- G. Water moves into the ground.
- H. Underground water flows into oceans.

25 Step A shows one way water can move into the air before it falls as rain.

Which statement **best** explains this process?

- A** Gravity changes water from liquid to gas.
- B** Gravity changes water from gas to liquid.
- C** The Sun's energy changes water from liquid to gas.
- D** The Sun's energy changes water from gas to liquid.

Standard: Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact.

Cognitive Complexity:

SEP: Med

DCI: Med

CCC: Med

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may think that gravity is responsible for evaporation.
- B. The student may think that gravity is responsible for evaporation and may not understand that evaporation changes water from a liquid to a gas.
- C. KEY: Heat from the Sun evaporates water from liquid into gas.**
- D. The student may not understand that evaporation changes water from a liquid to a gas.

26 Which model **best** shows how water moves between Earth systems to create rainfall on an island?

- A** atmosphere → hydrosphere → geosphere
- B** hydrosphere → atmosphere → geosphere
- C** biosphere → geosphere → hydrosphere
- D** geosphere → biosphere → hydrosphere

Standard: Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact.

Cognitive Complexity:

SEP: Low

DCI: Med

CCC: High

Overall Cognitive Complexity: Low Support

Rationale:

- A. The student may not understand that rain falls from the atmosphere to the geosphere.
- B. KEY: Water from ocean evaporation (hydrosphere to atmosphere) condenses and falls as rain on the island (atmosphere to geosphere).**
- C. The student may not understand that water will need to move through the atmosphere for it to rain.
- D. The student may be confused with how many plants get their water from the geosphere.

27 Step F of the water cycle model represents which movement between Earth systems?

- A** Water moves as a gas from the biosphere to the atmosphere.
- B** Water moves as a gas from the hydrosphere to the biosphere.
- C** Water moves as a liquid from the biosphere to the geosphere.
- D** Water moves as a liquid from the geosphere to the hydrosphere.

Standard: Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact.

Cognitive Complexity:

SEP: Med

DCI: Low

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. KEY: Water transpires as a gas from plants (biosphere) into the atmosphere.**
- B. The student may not understand the direction that water is moving in Step F.
- C. The student may not understand that water moves as a gas in Step F, and that it is moving into the atmosphere.
- D. The student may not understand that water moves as a gas in Step F, and may be confused with the process by which many trees get their water from the ground.

Sample Cluster for 5.ESS2.2:

Use the information to answer the following questions.

A student is on a field trip at the beach and looks out at the ocean. The student can't see where the ocean ends and wonders how much water is in the ocean compared to other water sources around the world. The student finds an incomplete table that lists the volume in thousands of cubic kilometers (km^3) of Earth's water sources.

Earth's Water Volume

Water Source	Volume (thousands of km^3)
Ocean	1,338,000
Groundwater	?
Lakes, rivers, streams, wetlands	?
Ice caps, glaciers, permanent snow	24,064

(Items on the following pages)

28 Which statement could the student make to **best** describe freshwater sources on Earth?

- A** Most of Earth's fresh water is in lakes, rivers, streams, and wetlands.
- B** There is a smaller fraction of Earth's fresh water in the ocean than in groundwater.
- C** Most of Earth's fresh water is in ice caps, glaciers, permanent snow, and groundwater.
- D** There is a smaller fraction of Earth's fresh water in ice caps, glaciers, and permanent snow than in other freshwater sources.

Standard: Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth.

Cognitive Complexity:

SEP: N/A

DCI: Med

CCC: Low

Overall Cognitive Complexity: Moderate Support

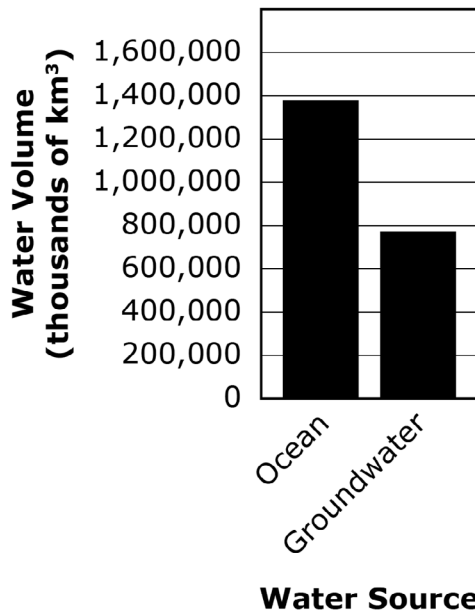
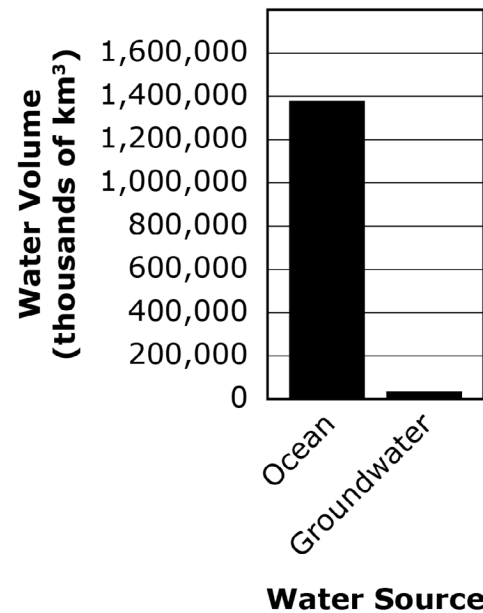
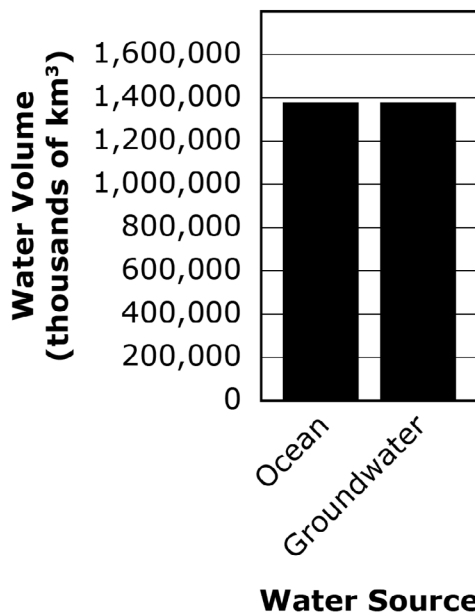
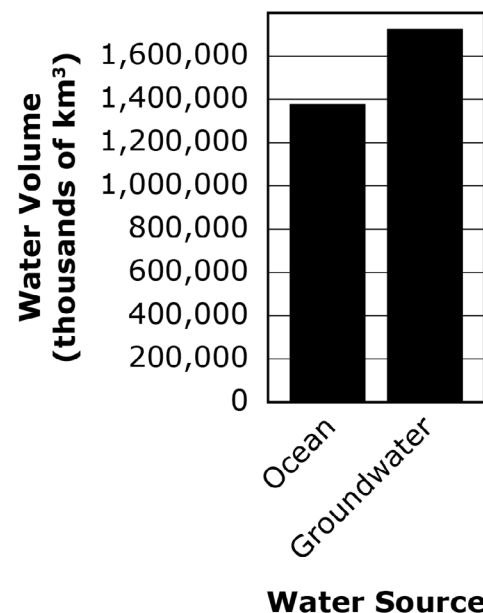
Rationale:

- A. Most of Earth's fresh water is in glaciers or groundwater.
- B. Students may have the misconception that the ocean is fresh water rather than salt water and that most fresh water is in lakes, rivers, streams, wetlands, glaciers, ice caps, and permanent snow instead of groundwater.
- C. KEY: This represents the correct distribution of Earth's fresh water, with most of it in glaciers and groundwater and the rest, a small fraction, in lakes, rivers, streams, and wetlands.**
- D. Students may not recognize that most fresh water is in ice caps, glaciers, and permanent snow rather than in other freshwater sources.

29

Based on the Earth's Water Volume table, the student wonders how the amount of groundwater on Earth compares to the amount of water in the ocean.

Which bar graph represents Earth's water sources by comparing the total volume of groundwater to the total volume of Earth's ocean?

A**Earth's Water Sources****B****Earth's Water Sources****C****Earth's Water Sources****D****Earth's Water Sources**

Standard: Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth.

Cognitive Complexity:

SEP: High

DCI: Med

CCC: Med

Overall Cognitive Complexity: Low Support

Rationale:

- A. Groundwater is a large freshwater source but is not so large that it would be half as much as the volume of ocean water since fresh water is still a much smaller fraction of Earth's total water volume.
- B. KEY: Groundwater is a much smaller fraction of Earth's total water volume compared to the ocean, since most of the water on Earth is salt water (the ocean).**
- C. Groundwater is a freshwater source and is therefore a much smaller fraction of Earth's total water volume than the ocean, so those water volumes are not equal.
- D. The ocean makes up the vast majority of Earth's total water volume, so groundwater cannot have a larger water volume than ocean water.

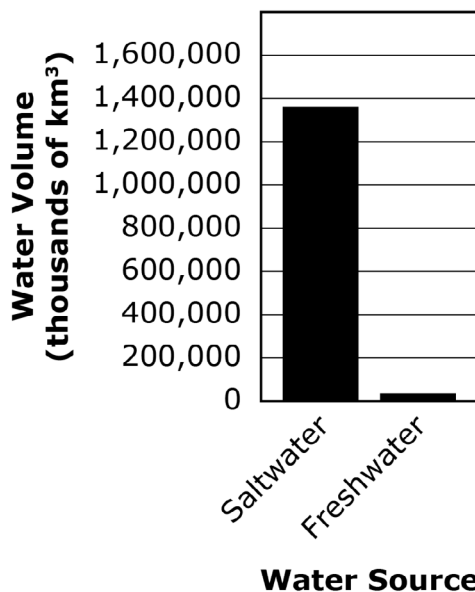
30

Based on the Earth's Water Volume table, the student wonders how the amount of salt water on Earth compares with the amount of fresh water.

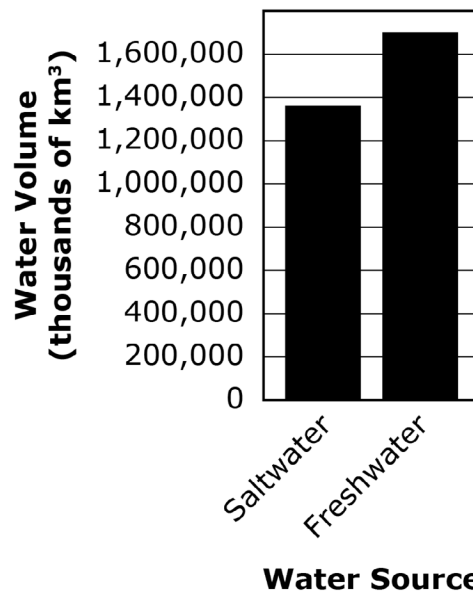
Which bar graph represents the comparison of saltwater and freshwater sources on Earth?

A

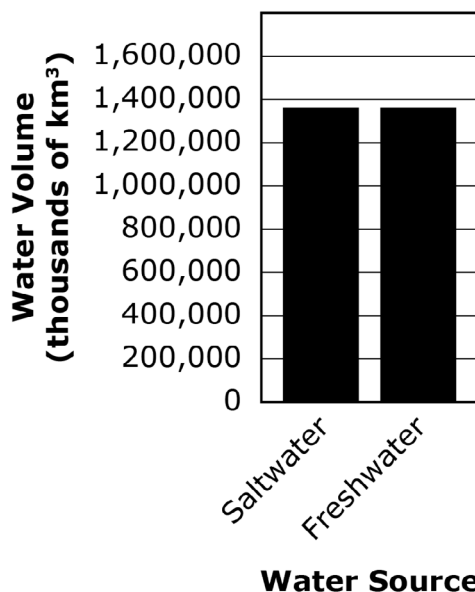
**Earth's Total
Saltwater and
Freshwater Sources**

**B**

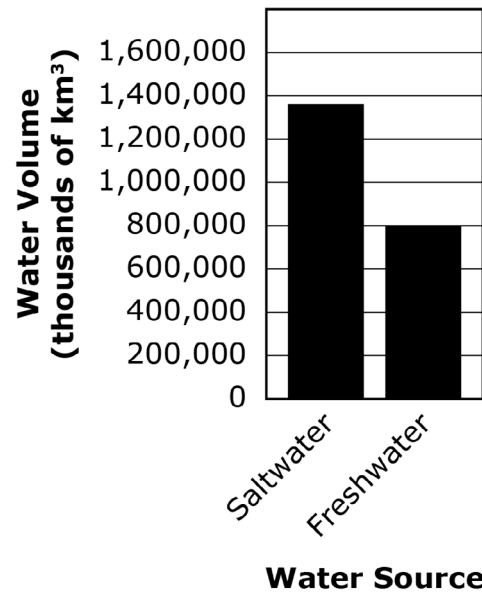
**Earth's Total
Saltwater and
Freshwater Sources**

**C**

**Earth's Total
Saltwater and
Freshwater Sources**

**D**

**Earth's Total
Saltwater and
Freshwater Sources**



Standard: Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth.

Cognitive Complexity:

SEP: High

DCI: Med

CCC: Med

Overall Cognitive Complexity: Low Support

Rationale:

- A. **KEY:** The ocean makes up the vast majority of Earth's water and consists of salt water, so there is much more salt water than fresh water on Earth.
- B. Students may think that the ocean is fresh water rather than salt water.
- C. Students do not recognize that there is much more salt water on Earth (the ocean) rather than fresh water.
- D. There is more salt water than fresh water on Earth but fresh water is a very small fraction, much smaller than half the amount of Earth's total saltwater volume.